**14. Fabricating Systems** A production plant has two fabricating systems, both of which are maintained at 2-week intervals. However, one system is twice as old as the other. The number of finished products fabricated daily by each of the systems is recorded for 30 working days, with the results given in the table. Do these data present sufficient evidence to conclude that the variability in daily production warrants increased maintenance of the older fabricating system? Use the $p$-value approach.

| New System | Old System |
|---|---|
| $\bar{x}_1 = 246$ | $\bar{x}_2 = 240$ |
| $s_1 = 15.6$ | $s_2 = 28.2$ |

$H_0 : \sigma_2^2 = \sigma_1^2$

$H_a : \sigma_2^2 > \sigma_1^2$

$F = \frac{s_2^2}{s_1^2} = \frac{28.2^2}{15.6^2} = \frac{795.24}{243.36} \approx 3.268$

$df_1 = n_2 - 1 = 29$

$df_2 = n_1 - 1 = 29$

P-value $= P(F_{29,29} > 3.268) < 0.001$

$\therefore p < 0.05$, Reject $H_0$. #

**16. SAT Scores** The SAT subject tests in chemistry and physics for two groups of 15 students each electing to take these tests are given here.

| Chemistry | Physics |
|---|---|
| $\bar{x} = 784$ | $\bar{x} = 758$ |
| $s = 114$ | $s = 103$ |
| $n = 15$ | $n = 15$ |

To use the two-sample $t$-test with a pooled estimate of $\sigma^2$, you must assume that the two population variances are equal. Test this assumption using the $F$-test for equality of variances. What is the approximate $p$-value for the test?

$H_0 : \sigma_1^2 = \sigma_2^2$

$H_a : \sigma_1^2 \neq \sigma_2^2$

$F = \frac{114^2}{103^2} = \frac{12996}{10609} \approx 1.225$

$df_1 = 14$, $df_2 = 14$ $\quad F_{14,14,0.10} = 2.01$

P-value $= 2 \times P(F_{14,14} > 1.225)$

$> 2 \times 0.10$

$> 0.20$

$\because F = 1.225 < 2.01$, $P(F > 1.225) > 0.10$, pvalue $= 2 \times P(F \geq 1.225) > 0.20$

$\therefore$ Non-reject $H_0$

**22. Stock Risks** The closing prices of two common stocks were recorded for a period of 15 days. The means and variances are

$\bar{x}_1 = 40.33$ $\quad$ $\bar{x}_2 = 42.54$

$s_1^2 = 1.54$ $\quad$ $s_2^2 = 2.96$

**a.** Do these data present sufficient evidence to indicate a difference between the variabilities of the closing prices of the two stocks for the populations associated with the two samples? Give the $p$-value for the test and interpret its value.

**b.** Construct a 99% confidence interval for the ratio of the two population variances.

a. $H_0: \sigma_1^2 = \sigma_2^2$

$H_a: \sigma_1^2 \neq \sigma_2^2$

$F = \frac{2.96}{1.54} \approx 1.922$

$df_1 = 14$ ; $df_2 = 14$ $\qquad$ $F_{14,14,0.10} = 2.01$

P-value $= 2 \cdot P(F_{14,14} \geq 1.922)$

$> 2 \times 0.10$

$> 0.20$

$\because F = 1.922 < 2.01$, $P(F \geq 1.922) > 0.10$, p-value $= 2 \cdot P(F \geq 1.922) > 0.20$

$\therefore$ Non-reject $H_0$.

b. The 99% CI for $\sigma_1^2/\sigma_2^2$ $= \left[\frac{s_1^2}{s_2^2} \frac{1}{F_{df_1,df_2,\frac{\alpha}{2}}} ; \frac{s_1^2}{s_2^2} F_{df_2,df_1,\frac{\alpha}{2}}\right]$

$= \left[\frac{1.54}{2.96} \frac{1}{F_{14,14,0.005}} , \frac{1.54}{2.96} F_{14,14,0.005}\right]$

$= \left[\frac{1.54}{2.96} \frac{1}{4.25} , \frac{1.54}{2.96} \cdot 4.25\right]$

$= [0.122 , 2.211]$

$\therefore$ The 99% CI for $\sigma_1^2/\sigma_2^2 = (0.122, 2.211)$ include 1, so Non-reject $H_0$.

DATA SET DS1040

**11. Reaction Times** A comparison of reaction times (in seconds) for two different stimuli in an experiment produced the following results when applied to a random sample of 16 people:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Stimulus 1 | 1 | 3 | 2 | 1 | 2 | 1 | 3 | 2 |
| Stimulus 2 | 4 | 2 | 3 | 3 | 1 | 2 | 3 | 3 |

Do the data present sufficient evidence to indicate a difference in mean reaction times for the two stimuli? Test using $\alpha = .05$.

$d(x_1 - x_2)$ $\quad$ -3 $\quad$ 1 $\quad$ -1 $\quad$ -2 $\quad$ 1 $\quad$ -1 $\quad$ 0 $\quad$ -1

$\bar{d} = \frac{-6}{8} = -0.75$

$H_0: \mu_d = 0$

$H_a: \mu_d \neq 0$

$S_d^2 = \frac{1}{n-1}\left[\sum_{i=1}^{n} d_i^2 - n\bar{d}^2\right]$

$= \frac{1}{7}[(9+1+1+4+1+1+1) - [8 \times (-0.75)^2]$

$= \frac{1}{7}(19.5)$

$= 1.9286$

$S_d = \sqrt{1.9286}$

$= 1.389$

$t_{df,\frac{\alpha}{2}} = t_{7,0.025}$

$|t| = 2.3646$

$t = \frac{\bar{d} - \mu_0}{S_d/\sqrt{n}} = \frac{-0.75}{1.389/\sqrt{8}}$

$= -1.53$

$df = n - 1 = 7$

RR $= \{T > 2.3646\}$

$\because |-1.53| = 1.53 < 2.3646$

$\therefore$ Non-reject $H_0$

DATA SET DS1041

**12. Reaction Times II** Refer to Exercise 11. Suppose that the experiment is conducted using eight people as blocks and making a comparison of reaction times within each person; that is, each person is subjected to both stimuli in a random order. The reaction times (in seconds) for the experiment are as follows:

| Person | Stimulus 1 | Stimulus 2 | |
|---|---|---|---|
| 1 | 3 | 4 | -1 |
| 2 | 1 | 2 | -1 |
| 3 | 1 | 3 | -2 |
| 4 | 2 | 1 | 1 |
| 5 | 1 | 2 | -1 |
| 6 | 2 | 3 | -1 |
| 7 | 3 | 3 | 0 |
| 8 | 2 | 3 | -1 |

Do the data present sufficient evidence to indicate a difference in mean reaction times for the two stimuli? Test using $\alpha = .05$.

$\bar{d} = \frac{-6}{8} = -0.75$

$H_0: \mu_d = 0$

$H_a: \mu_d \neq 0$

$S_d^2 = \frac{1}{n-1}\left[\sum_{i=1}^{n} (d_i^2 - n\bar{d}^2)\right]$

$= \frac{1}{7}\left[10 - (8 \times (-0.75)^2)\right]$

$= 0.7857$

$S_d = \sqrt{0.7857}$

$= 0.886$

$t = \frac{\bar{d} - \mu_0}{S_d/\sqrt{n}}$

$= \frac{-0.75}{0.886/\sqrt{8}}$

$= -2.394$

$df = n-1 = 7$

$t_{df, \frac{\alpha}{2}} = t_{7, 0.025}$

$|t| = 2.3646$

$RR = \{T > 2.3646\}$

$\because |-2.394| = 2.394 > 2.3646$

$\therefore$ Reject $H_0$.